

Supplementary Information: Validated Synthetic Patient Generation for Small Longitudinal Cohorts

Supplementary Figures

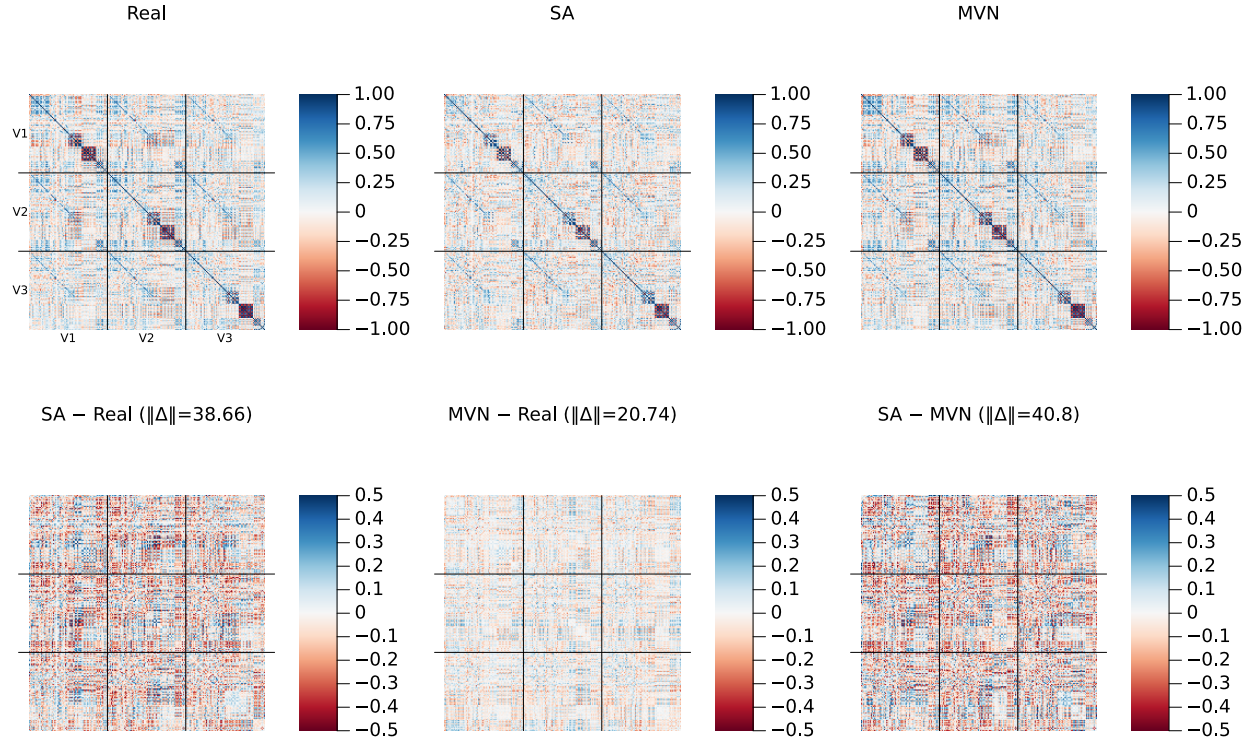


Figure S1: **Cross-visit correlation structure (amplified residual scale)**. Same layout as the main-text figure, but with the bottom-row residual panels plotted on a ± 0.5 color scale (vs. ± 1.0 in the main text) to reveal fine structure in the correlation differences. Frobenius norms of each residual matrix are shown in the panel titles. The MVN–Real residual is systematically smooth (reflecting regularization-induced shrinkage), while the SA–Real residual is noisier but unbiased.

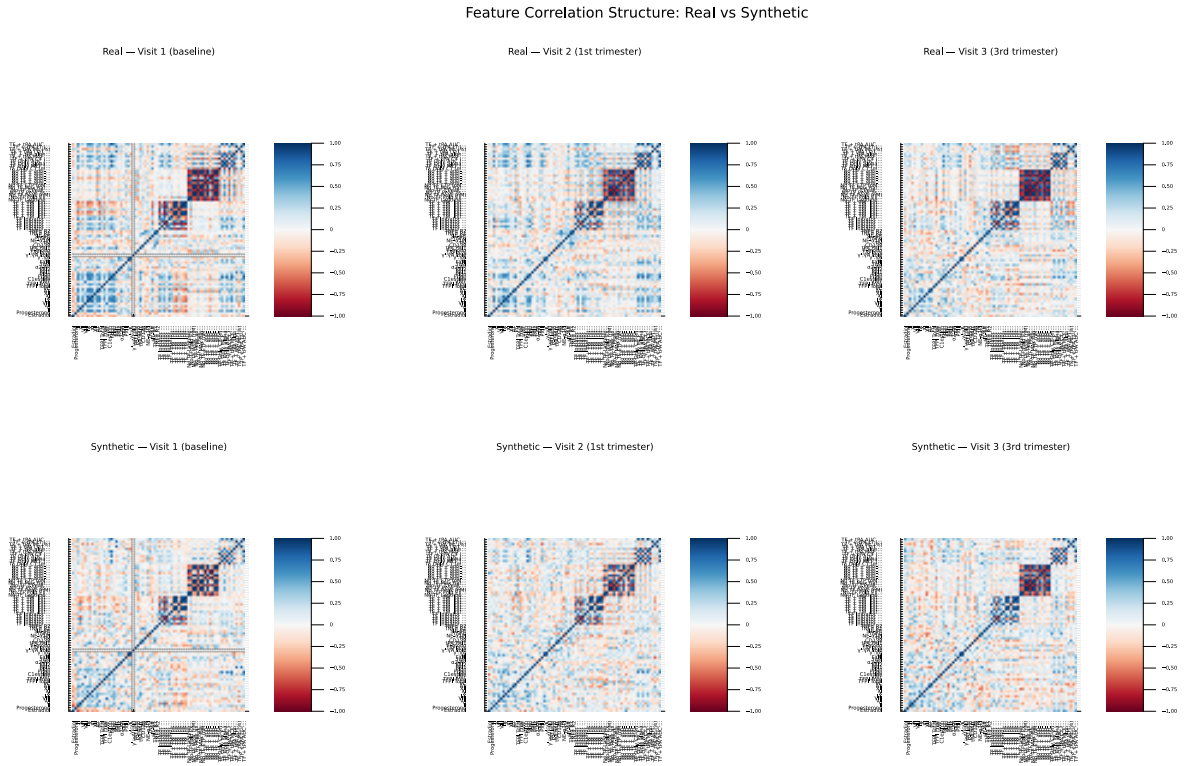


Figure S2: **Within-visit correlation heatmaps.** Pearson correlation matrices for real and SA-generated synthetic patients within each visit (72 features per visit). SA preserves the within-visit correlation structure across all three visits.

Supplementary Tables

Supplementary Figures

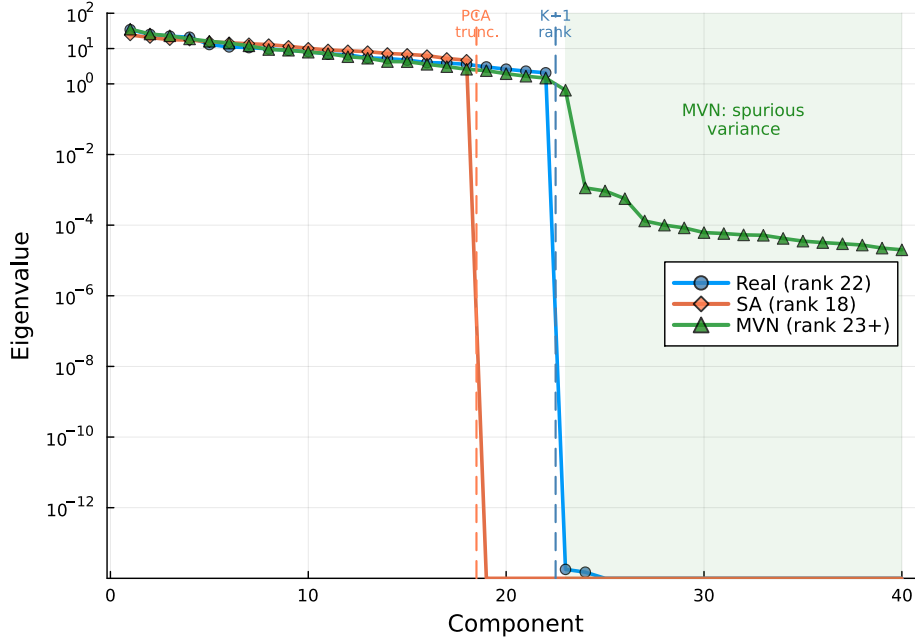


Figure S3: **Covariance eigenvalue spectrum.** Log-scale eigenvalues of the 216×216 sample covariance for Real ($K=23$, rank 22), SA ($N=100$, rank 18), and MVN ($N=100$) populations. Dashed vertical lines mark the SA PCA truncation point (component 18) and the data rank limit (component 22, $K-1$). SA’s spectrum drops cleanly at the PCA truncation, matching the low-rank data structure. MVN’s Ledoit–Wolf regularization inflates eigenvalues beyond component 22 (green shaded region), introducing spurious variance in 194 dimensions where the data contain none. This spurious variance is the structural origin of the increased MVN dispersion visible in the PCA projections (main-text Fig. 4).

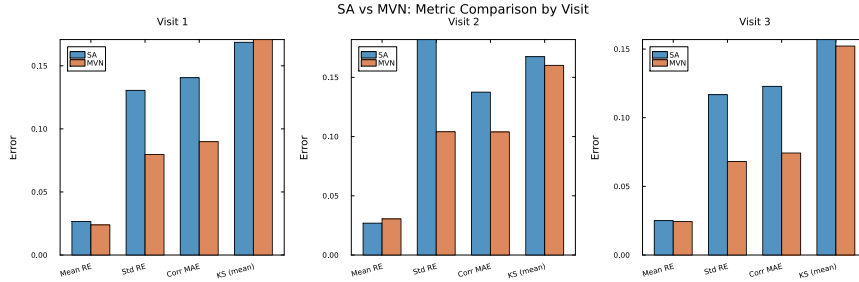


Figure S4: **SA vs. MVN quantitative comparison.** Per-visit comparison of mean relative error, standard deviation relative error, KS statistic, and correlation MAE between SA and MVN generation methods. Both methods perform comparably on marginal statistics (Level 1), but diverge on structural measures (correlation MAE).

Table S1: **Inverse temperature sensitivity analysis.** SA generation quality as a function of β/β^* , where $\beta^* \approx 2.94$ was identified via entropy inflection in the 18-dimensional PCA space ($K=23$ stored patterns). The theoretical prediction $\beta^* = \sqrt{d} = \sqrt{18} \approx 4.24$ is close to the empirical value. Operating at $\beta = \beta^*$ (bold row) balances generation novelty against fidelity: lower β produces more diverse but less faithful samples; higher β produces more faithful but less novel samples (approaching memorization). MRE = median relative error of per-feature means; Corr MAE = mean absolute error of the correlation matrix; Novelty = $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$.

β/β^*	β	Noise scale	PC1 std ratio	Novelty	Med MRE (%)
0.1	0.29	0.261	0.538	0.542	0.7
0.2	0.59	0.184	0.545	0.539	0.7
0.3	0.88	0.151	0.550	0.536	0.7
0.5	1.47	0.117	0.562	0.528	0.7
0.7	2.06	0.099	0.573	0.518	0.7
1.0	2.94	0.082	0.589	0.502	0.6
1.5	4.41	0.067	0.613	0.474	0.7
2.0	5.88	0.058	0.638	0.446	0.7
3.0	8.82	0.048	0.689	0.403	0.8

Table S2: **Multiplicity weighting parameters for conditional generation.** For each clinical subgroup, the multiplicity ratio ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated patterns. K_{eff} is the participation ratio, quantifying the effective number of patterns contributing to generation. PCOS required the highest ρ (26.7) due to having only 3 stored patterns, reducing K_{eff} to 4.6 and explaining the larger MREs observed for this subgroup at Level 3.

Condition	K_{sub}	K_{bg}	ρ	f_{eff}	K_{eff}
Healthy	14	9	2.57	0.80	19.9
PCOS	3	20	26.67	0.80	4.6
PE	5	18	14.40	0.80	7.7

Table S3: **PCA variance threshold sensitivity analysis.** SA generation quality as a function of the PCA variance retention threshold (d_{PCA} ranges from 13 to 21 dimensions). The 95% threshold used in the main text (bold) is not a critical choice: median MRE is stable at 1.2–1.8% across all thresholds, and correlation MAE varies modestly (0.116–0.148). Higher thresholds retain more variance but reduce the K/d ratio, approaching the regime where SA has fewer stored patterns than dimensions. Novelty increases with d_{PCA} because more dimensions provide more room for interpolation.

Threshold	d_{PCA}	K/d	β^*	Novelty	Med MRE	Corr MAE
85%	13	1.77	2.94	0.413	0.012	0.116
90%	15	1.53	2.94	0.462	0.018	0.131
95%	18	1.28	2.94	0.516	0.012	0.148
98%	20	1.15	2.94	0.538	0.013	0.127
99%	21	1.10	2.94	0.540	0.013	0.133

Table S4: **Comparison with deep generative baselines.** CTGAN and TVAE (from the Synthetic Data Vault library) were trained on the same 90 per-visit records (23 patients $\times$ 3 visits, 72 features) at three epoch counts. SA and MVN results from the main text are shown for reference. CTGAN fails catastrophically at this sample size (median MRE $\approx$ 19%, 10 $\times$ worse than SA), consistent with known GAN mode collapse behavior on small datasets. TVAE achieves comparable marginal fidelity to SA at high epoch counts (MRE 1.8% at 3,000 epochs) but operates on per-visit rows rather than concatenated longitudinal profiles, so it cannot capture cross-visit covariance structure and has no mechanism for conditional subgroup generation. Med KS = median two-sample KS statistic across all features; Corr MAE = mean absolute error of the within-visit correlation matrix.

Method	Epochs	Med MRE	Med KS	Corr MAE
SA	—	0.019	0.169	0.200
MVN	—	0.024	0.161	0.090
CTGAN	300	0.189	0.364	0.261
CTGAN	1,000	0.195	0.349	0.265
CTGAN	3,000	0.187	0.341	0.180
TVAE	300	0.037	0.174	0.152
TVAE	1,000	0.026	0.248	0.082
TVAE	3,000	0.018	0.224	0.092

Table S5: **Summary of mean relative errors across all 72 features.** Per-visit distribution of MREs between real and SA-generated synthetic patient means. The full feature-by-feature table (72 features $\times$ 3 visits = 216 entries) is provided in the code repository as `paper_summary_statistics.csv`.

Visit	Median MRE	Mean MRE	25th pctl	75th pctl	Max MRE
V1 (baseline)	0.019	0.027	0.010	0.033	0.157
V2 (1st tri)	0.019	0.027	0.009	0.034	0.124
V3 (3rd tri)	0.016	0.025	0.006	0.040	0.110

Table S6: **SA generation hyperparameters.** All parameters used for synthetic patient generation. The same values were used for unconditioned and conditioned generation except where noted.

Parameter	Value	Description
α	0.01	Langevin step size (ULA valid without MH correction)
T	2,000	Langevin iterations per sample
β^*	2.94	Inverse temperature (entropy inflection)
PCA threshold	95%	Variance retained
d_{PCA}	18	PCA dimensionality
N	100	Synthetic patients per cohort
Random seed	42	For reproducibility
f_{target}	0.80	Target attention fraction (conditioned only)

Table S7: **BZ2012 ODE model calibration.** Five rate constants were calibrated on Visit 1 real patients ($n=23$) using bounded Nelder–Mead optimization (Fminbox) in log-scale-factor space. The cost function was the mean normalized relative squared error across five TGA features (lagtime, peak, time-to-peak, maximum rate, ETP) under both TF-only and TF+TM conditions. Bounds: $\pm \log(100)$ ($0.01\times$ to $100\times$ default). Convergence: $f_{\text{tol}} = 10^{-6}$, 500 iterations.

Rate constant	Role	Default	Calibrated	Scale
prothrombinase k_{cat}	Thrombin burst	63.5 s^{-1}	21.94 s^{-1}	$0.35\times$
intrinsic Xase k_{cat}	Amplification Xa	8.2 s^{-1}	0.169 s^{-1}	$0.021\times$
extrinsic Xase k_{cat}	Initiation Xa	6.0 s^{-1}	93.2 s^{-1}	$15.5\times$
PC activation k_{cat}	Protein C activation	0.41 s^{-1}	0.890 s^{-1}	$2.17\times$
mIIa conversion k	mIIa $\rightarrow$ IIa	2.3×10^8	1.15×10^9	$5.01\times$

Units for mIIa conversion: $\text{M}^{-1}\text{s}^{-1}$

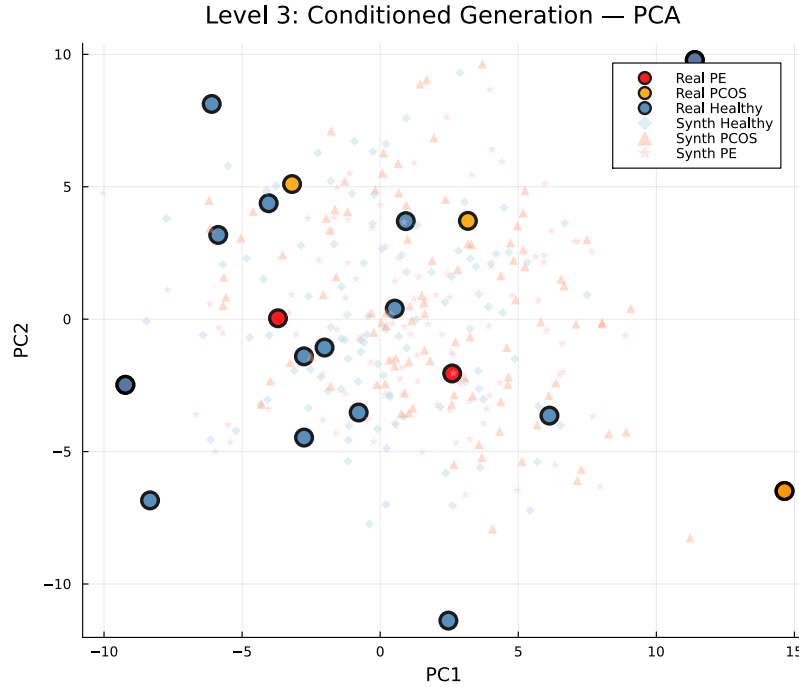


Figure S5: **Conditioned generation: PCA projection (pooled).** PCA projection of condition-specific synthetic cohorts generated by multiplicity-weighted SA. Large markers: real patients (blue = Healthy, $n=14$; orange = PCOS, $n=3$; red = PE, $n=5$). Small markers: synthetic patients ($N=100$ per condition). With only 3 PCOS and 5 PE patients, the subgroups do not form clearly separable clusters in the first two principal components, but the synthetic cohorts concentrate around their respective real counterparts.

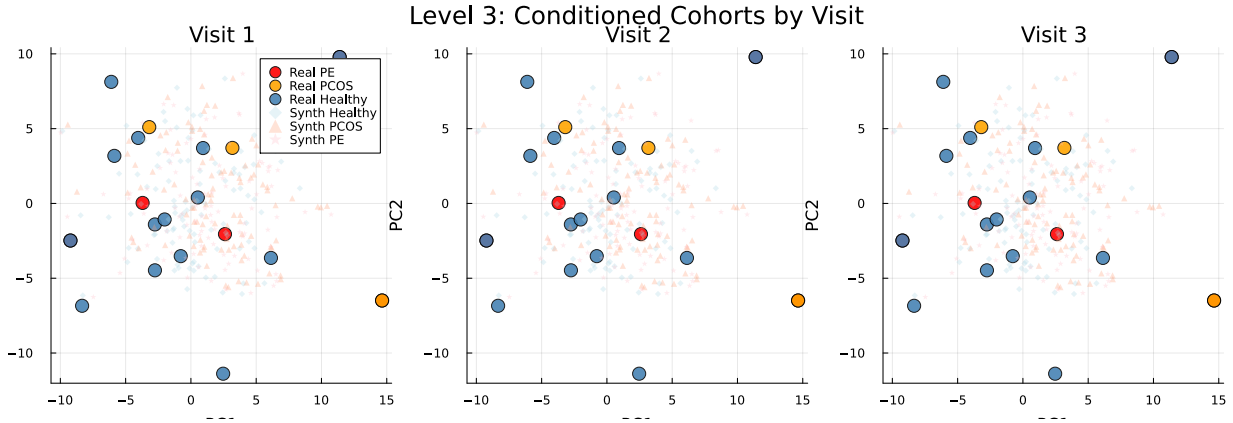


Figure S6: **Conditioned generation: PCA projection by visit.** Same as Supplementary Fig. S5 but separated by visit (V1, V2, V3). The condition-specific clustering patterns are maintained within each individual visit.

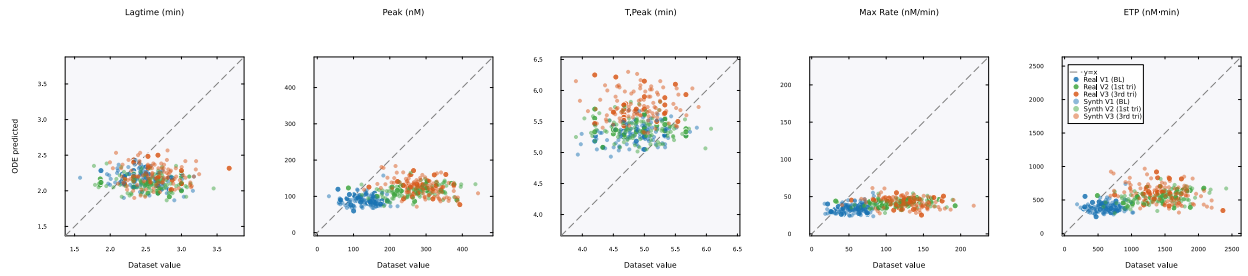


Figure S7: **Level 4: Mechanistic model predicted vs. measured TGA features (TF+TM condition).** Same layout as main-text Fig. 6. The model systematically underpredicts peak and maximum rate ($\approx 0.5\times$ measured) for both real and synthetic patients, reflecting overcorrection of the protein C anticoagulant pathway. The shared bias indicates that synthetic patients are processed identically to real patients by the mechanistic model.

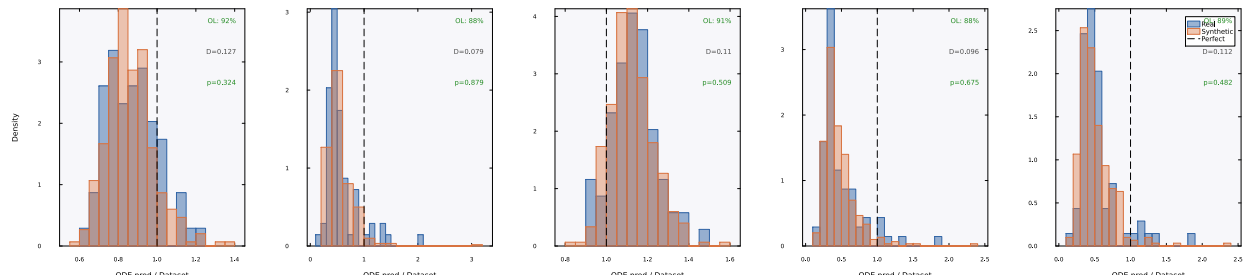


Figure S8: **Level 4: Predicted/measured ratio distributions (TF+TM condition).** Same layout as main-text Fig. 7. Despite the systematic bias in peak and maximum rate, the ratio distributions for real and synthetic patients overlap substantially (88–91% cloud overlap).

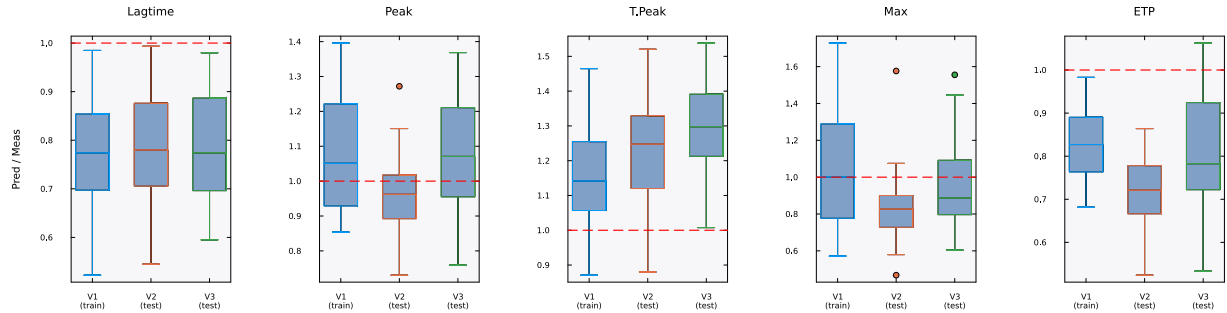


Figure S9: **Level 4: Cross-visit calibration generalization (TF-only).** Boxplots of the predicted/measured ratio by visit for real patients. The model was calibrated on Visit 1 (“train”) only. Ratios near 1.0 at Visits 2 and 3 indicate that the population-level calibration generalizes across pregnancy timepoints.

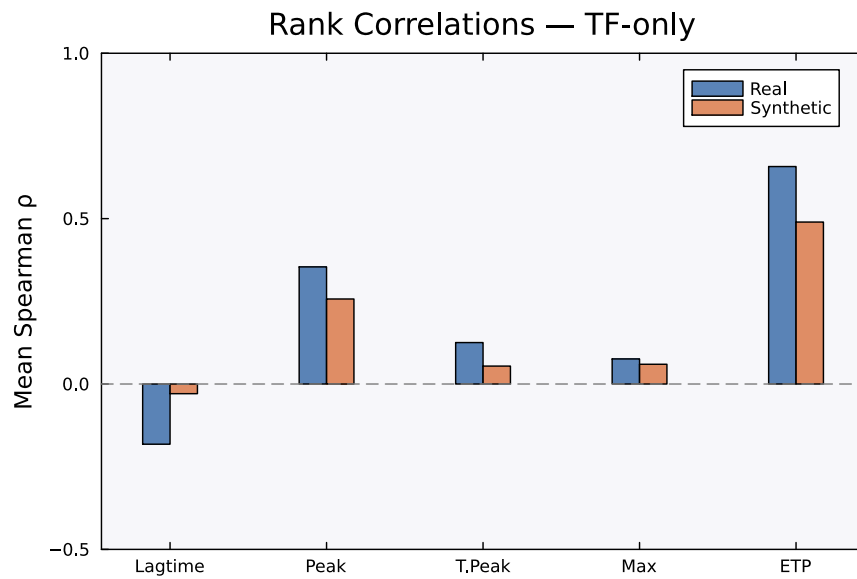


Figure S10: **Level 4: Spearman rank correlations (TF-only).** Mean Spearman ρ between predicted and measured TGA features, averaged across visits, for real and synthetic patients. ETP is the best-predicted feature ($\rho \approx 0.6\text{--}0.8$). Weak rank correlations for other features are expected given the population-level (not patient-level) calibration.

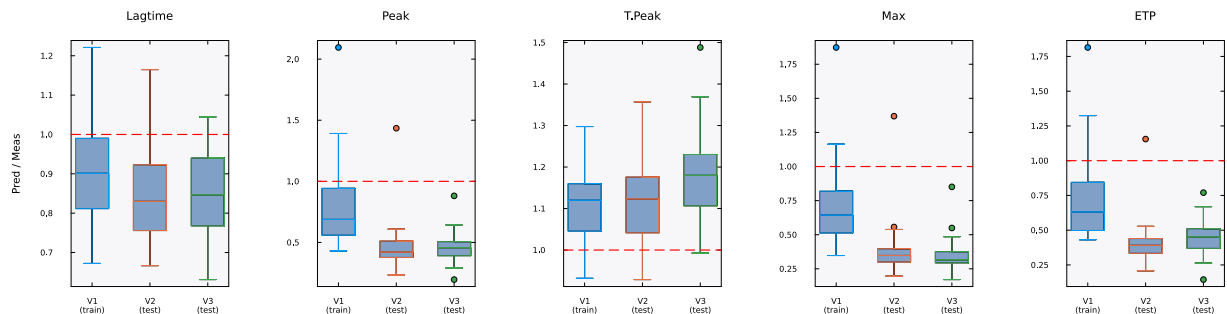


Figure S11: **Level 4: Cross-visit calibration generalization (TF+TM).** Same layout as Supplementary Fig. S9. The TF+TM calibration does not generalize as well as TF-only, particularly for peak and maximum rate, due to overcorrection of the protein C pathway.

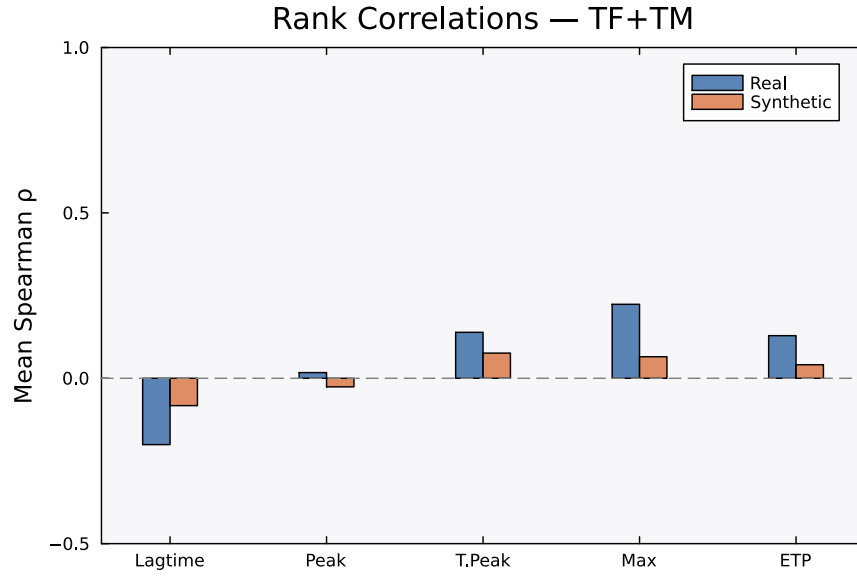


Figure S12: **Level 4: Spearman rank correlations (TF+TM)**. Same layout as Supplementary Fig. S10. Rank correlations under TF+TM are generally weaker than TF-only, consistent with the poorer calibration generalization under this condition.

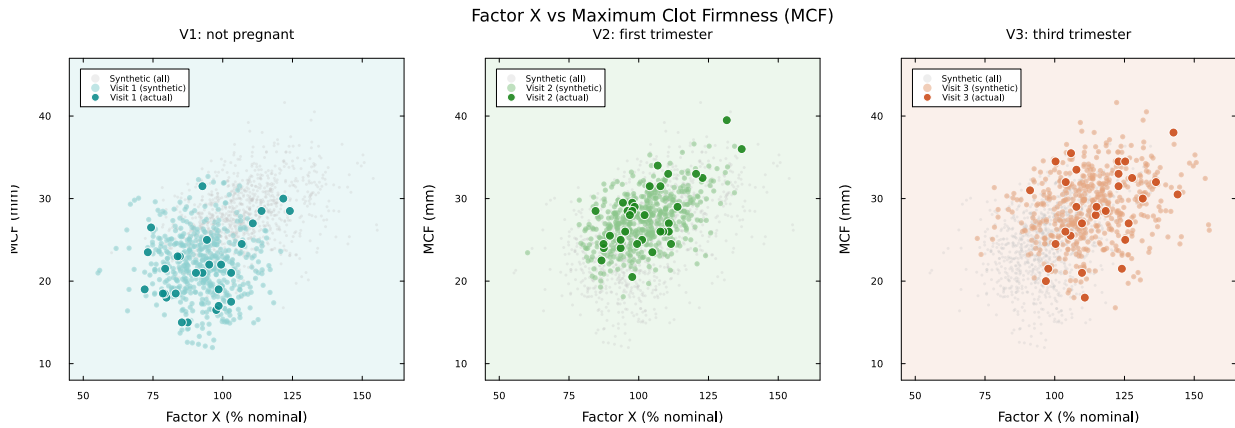


Figure S13: **Factor X vs. Maximum Clot Firmness (MCF) trajectory**. The characteristic pregnancy-driven shift in the Factor X–MCF plane is preserved in SA-generated synthetic patients, demonstrating that biologically meaningful multivariate trajectories are captured by the generation pipeline.

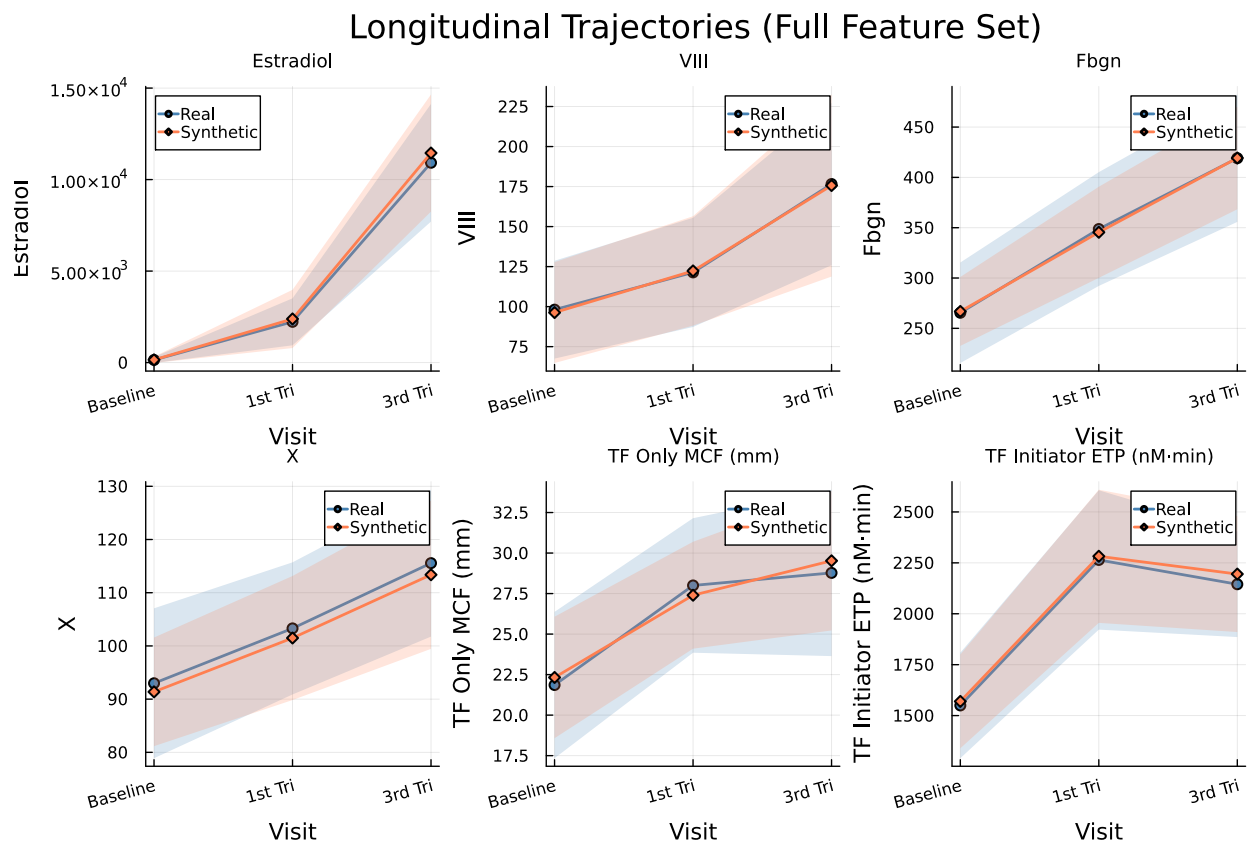


Figure S14: **Full longitudinal trajectories.** Individual patient trajectories (all 72 features) across three visits for both real and SA-generated synthetic patients, demonstrating coherent longitudinal patterns.